



AI call centers

Google's AI assistant demonstration was amazing at I/O 2018, and the company might deploy Duplex into call centers. <https://dgit.in/Aug18-19-1>

nounced earlier came into being by 2005, around the same time, Samsung announced the development of a 2GB NAND flash storage. The price of NAND flash had already dropped below DRAM for the same density. Hynix and ST Micro formed a joint venture and Hynix NAND product was introduced. Infineon NAND product based on Saifun Charge Trap Flash was announced. But even with all these developments, bigger upheavals were headed towards NAND memory.



The first gen iPhone holds significant credit for generating a huge demand and popularity for NAND flash

FLASH AS WE KNOW IT

On January 9, 2007, the iPhone was announced by Steve Jobs at the Macworld convention. Yes, it did come with NAND flash storage, as have many Android-powered and other smartphones. The growing popularity of iPhones, along with categories like iPods that were popular at the time, the demand for NAND chips soared. In 2009, Apple signed a long term contract with Toshiba for NAND chips, cementing future demand for NAND chips.

On the enterprise side of things, EMC (currently Dell EMC) intro-



Samsung started off V-NAND with its 850 Pro SSD

duced new 73GB and 146GB SSDs as options for their high-end Symmetrix DMX-4 systems. Additionally, while NAND flash memory was always heading towards smaller and smaller dimensions, it was in 2010 that Intel and Micron sampled 25 nm NAND products. Later that year, and next year, Samsung and Sandisk started shipping 27 nm and 24 nm NAND products respectively.

A startup in Salt Lake City, Utah was also trying something

different on the enterprise front at this point. Fusion.io was "trying to accelerate the other side of the internet", as Steve Wozniak (who eventually joined the company as Chief Scientist in 2009) would put it, by putting NAND flash on servers using PCIe.

Meanwhile, the way NAND chips are built was due for a change. Instead of building out, NAND architects began to build up, stacking chips vertically. This led to 3D NAND, or as Samsung calls it, V-NAND. 3D NAND allows greater density and better performance without an increase in size, while overcoming the interference and manufacturing challenges which had previously been limiting NAND technology. Further developments led to an increasing number of layers, greater capacities, better performance.

Quite recently, Samsung announced the mass production of its 5th gen V-NAND memory. According to Samsung, the new 256Gb or gigabit V-NAND flash memory has reached a speed of 1.4Gbps, or gigabits per second, resulting in a 40 percent increase from its 64-layer predecessor. From this point onwards, things can only get faster, smaller and more reliable. 



Enhance your Presentations



PresenTorchc9

Wireless Presenter with LED Flash Light Torch

Best IT World (India) Pvt. Ltd.: Email: sales@iBall.co.in
Mumbai - 022 - 6774 5100 / 3081 5100 / 98900 52890,
Ahmedabad - 9909012276, **Bangalore** - 9611901010,
Chennai - 42168560 / 61, **Cochin** - 9447771400, **Dehradun** -
 9639004150, **Delhi** - 9643302043, **Goa** - 8007035766,
Guwahati - 2460207 / 9435731459, **Haryana** - 7065588902,
Hubli - 9611194669, **Indore** - 9691004777, **Jaipur** -
 9351782703 / 9351749881, **Jammu** - 9419293112, **Kolkata** -
 8017518591, **Lucknow** - 7617004075, **Ludhiana** -
 8725047804, **Nagpur** - 2244074, **Nashik** - 9371469439,
Orissa - 9861033284, **Patna** - 9304678951, **Pune** -
 9765046663, **Raipur** - 9009989262, **Ranchi** - 9386655599,
Secunderabad - 27894847, **Vijaywada** - 9248233668

www.iBall.co.in | Over 65 million iBall products sold